Minimal Spanning Trees [MST]

Defn :- Let G be a weighted graph. The minimal spanning tree of G is a spanning tree from G with minimum weight.

[1] Kruskal's Algorithm for MST :-

Input : A weighted graph.

Output : A spanning tree with minimum weight.

Step-1 :- List the edges with weights in ascending order.

Step-2 :- Start with the vertex connecting the edge with minimum weight. This is the first vertex of the tree T and the minimum weight is the first edge. From this vertex go to the other vertex through the edge with minimum weight such that this does not form a loop or circuit. In case two minimum are available, select any one arbitrarily.

Step-3 :- Continue the step 2 until contains (n-1) edge, where n is the number of vertices.

[2] Prim's Algorithm for minimal spanning Tree :-

Input : A weighted graph G with n vertices.

Output : A minimal spanning Tree with (n-1) edges.

St-1 :- Remove all self loops and parallel edges except the edge which has minimum weight for all pairs of vertices

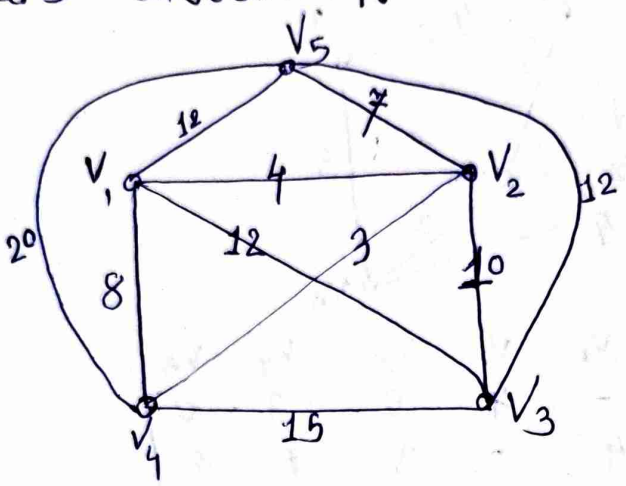
St-2 : Prepare weight table where
w_{ij} = weight bet. v_i & v_j (adj. vert.)
$w_{ij} = \infty$ if v_i & v_j not adjacent
& since no loops exists, put - as diagonal elements

St-3 :- Take the 1st row & find the min. weight in first row. Let it be ith ~~row~~ column. i.e. draw v_1 to v_i.

St-4 :- Now find smallest entry in first and ith row. Let it be jth row. Let the smallest entry be v_i. Add this to the subgraph.

St-5 :- Continue this process until all n vertices are connected by (n-1) edges. If there is a tie for selecting the smallest entry in any row then any one may be selected arbitrarily.

Ex-1: Draw MST of the graph by using Kruskal's algorithm.

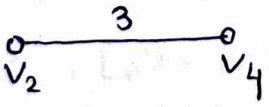


Soln Step-1 :- List the edges in ascending order.

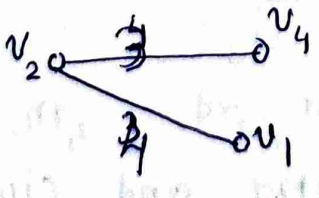
Edge	V_2-V_4	V_2-V_1	V_2-V_5	V_1-V_4	V_2-V_3	V_1-V_3
weight	3	4	7	8	10	12

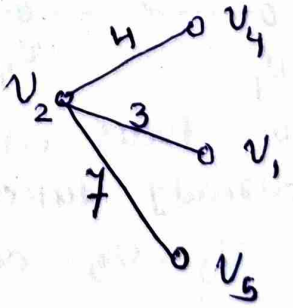
V_1-V_5	V_3-V_5	V_3-V_4	V_4-V_5
12	12	15	20

Step-2 :- Take min. weight 3



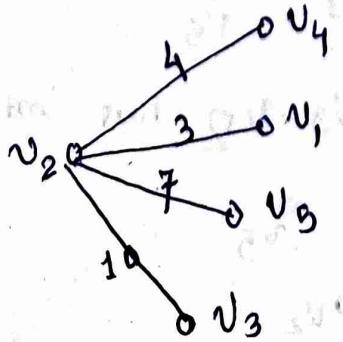
Then $V_2 \rightarrow V_1$; Then $V_2 \longrightarrow V_5$





{ V_1-V_4 Rejected as it's forming a circuit }

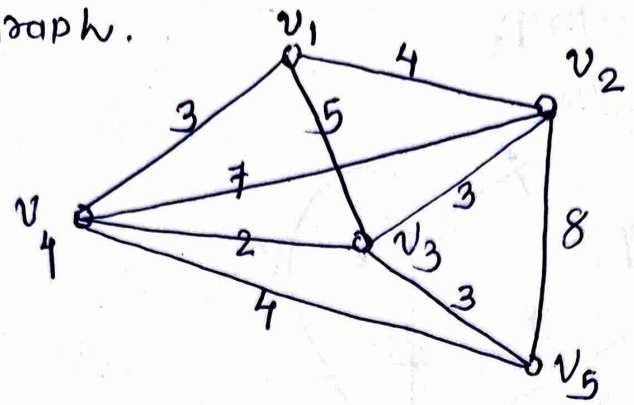
Then $V_2 \rightarrow V_3$



{ similarly V_1-V_3, V_3-V_5, V_3-V_4, V_4-V_5 Rejected }

← This is the Const. of minimal spanning tree & Minimum weight = 3+4+7+10 = 24

Ex-2: Find the minimum spanning Tree from, the graph. by prims ALG.

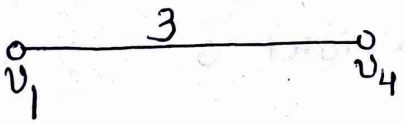


soln

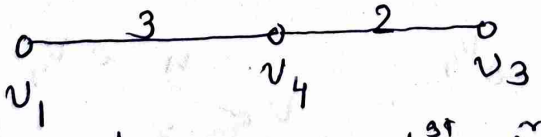
Let

$$W = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_3 & v_4 & v_5 \end{array} \\ \begin{bmatrix} - & 4 & 5 & 3 & \infty \\ 4 & - & 3 & 7 & 8 \\ 5 & 3 & - & 2 & 3 \\ 3 & 7 & 2 & - & 4 \\ \infty & 8 & 3 & 4 & - \end{bmatrix} \end{array}$$

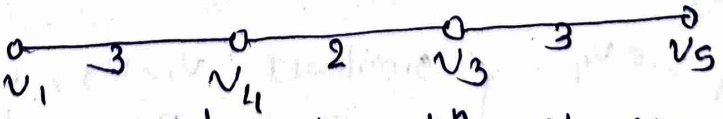
→ starting from v_1 & taking the lowest entry of row 1 which is 3 [$v_1 \leftrightarrow v_4$]



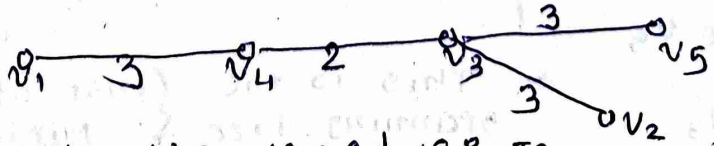
→ As min. weight occurs [v_1 - v_4], next selection will be from 1st & 4th row which is 2 [v_3 - v_4]



→ Next to focus on 1st, 3rd & 4th row [Reject already taken entry and circuit also] we've v_3 - v_5 as min = 3.



→ From 1st, 3rd, 4th, 5th v_3 - v_2 has min = 3

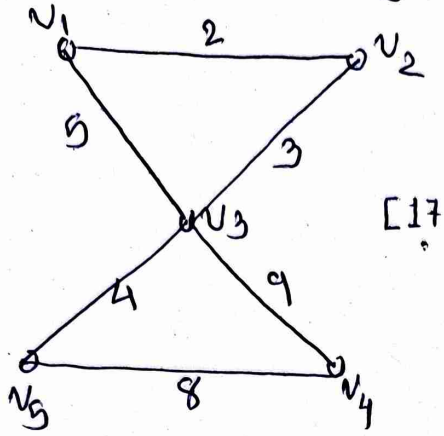


Above is the reqd. sp. Tree; min w. = 3 + 2 + 3 + 3 = $\boxed{11}$

Problems for finding MST

Q-1 Find MST by Kruskal's ALGO.

(1)



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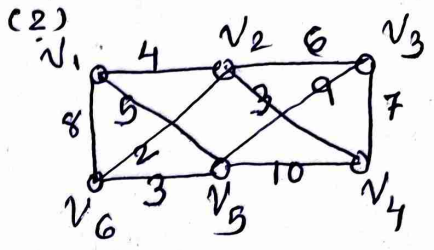
(2)



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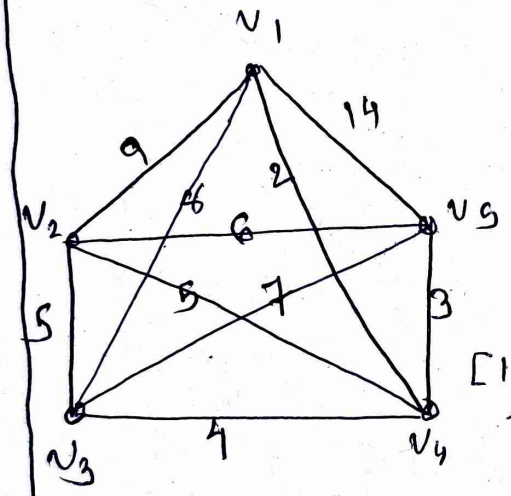
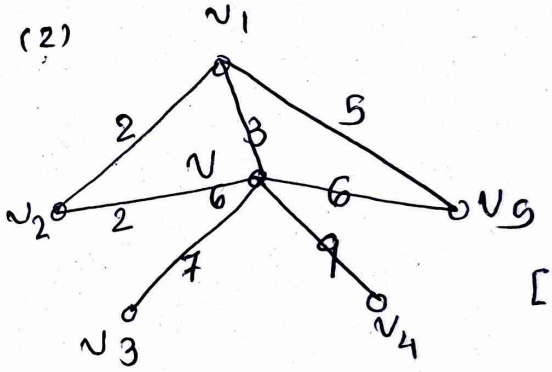
Q-2 Find MST by Prims ALGO.

(1)



[14]

(2)



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